

Complete Machine Learning Lab Viva Answers (BAIL606)

Experiment 1: Descriptive Statistics and Visualization

1. What is the difference between mean and median?

Mean is the average of all values, while median is the middle value after sorting.

2. Why is mode useful in data analysis?

Mode shows the most frequently occurring value.

3. What is standard deviation?

It measures how much values deviate from the mean.

4. Define variance.

Variance is the square of standard deviation.

5. What is range in statistics?

Range = Maximum value – Minimum value.

6. What is the purpose of a histogram?

Histogram shows frequency distribution of numerical data.

7. Why do we use a boxplot?

To visualize distribution, quartiles, and outliers.

8. What are outliers?

Outliers are values significantly different from others.

9. Explain IQR method for outlier detection.

Values below $Q1 - 1.5(IQR)$ or above $Q3 + 1.5(IQR)$ are outliers.

10. Difference between categorical and numerical data?

Categorical data are labels; numerical data are numbers.

11. Why are bar charts used for categorical variables?

To compare frequencies of categories.

12. Which Python libraries are used in this experiment?

NumPy, Pandas, Matplotlib, and Seaborn.

Experiment 2: Correlation and Covariance

1. What is correlation?

Correlation measures relationship between two variables.

2. What is Pearson correlation coefficient?

Measures strength and direction of linear relationship.

3. What is the range of correlation coefficient?

-1 to +1.

4. Difference between positive and negative correlation?

Positive means both increase together; negative means one decreases when other increases.

5. What is covariance?

Shows how variables vary together.

6. Difference between covariance and correlation?

Correlation gives both strength and direction; covariance gives direction only.

7. What does a scatter plot represent?

Relationship between two variables.

8. What is a correlation matrix?

Table showing correlations among variables.

9. Why is a heatmap used?

To visualize correlation values using colors.

10. What does a correlation value close to +1 indicate?

Strong positive relationship.

11. Can correlation imply causation?

No, correlation does not imply causation.

12. Which library is commonly used for heatmaps?

Seaborn.

Experiment 3: PCA

1. What is PCA?

PCA is a dimensionality reduction technique.

2. Why is dimensionality reduction needed?

To reduce complexity and improve performance.

3. What are principal components?

New variables formed from original features.

4. How does PCA reduce dimensions?

By selecting components with maximum variance.

5. What is variance in PCA?

Variance represents amount of information in data.

6. What are eigenvalues and eigenvectors?

Eigenvectors determine directions; eigenvalues determine importance.

7. Why is data normalization important before PCA?

Ensures equal contribution of features.

8. How many features are present in Iris dataset?

4 features.

9. What is the advantage of PCA?

Reduces dimensions while preserving information.

10. Can PCA remove noise from data?

Yes.

11. What is feature extraction?

Creating new meaningful features.

12. Mention real-world applications of PCA.

Face recognition, image compression, visualization.

Experiment 4: k-NN

1. What is k-NN algorithm?

A supervised algorithm used for classification and regression.

2. Is k-NN supervised or unsupervised?

Supervised learning.

3. What is the role of K value?

K determines number of nearest neighbors considered.

4. What happens if K=1?

Prediction depends on nearest single data point and may overfit.

5. What distance metric is commonly used in k-NN?

Euclidean distance.

6. What is weighted k-NN?

Closer neighbors are given more importance.

7. Difference between regular and weighted k-NN?

Weighted k-NN gives higher importance to closer neighbors.

8. What is overfitting in k-NN?

Model performs well on training data but poorly on new data.

9. Why do we split data into training and testing sets?

To evaluate performance on unseen data.

10. What is accuracy?

Ratio of correct predictions to total predictions.

11. What is F1-score?

Harmonic mean of precision and recall.

12. What are the disadvantages of k-NN?

Slow for large datasets and sensitive to noise.

Experiment 5: FIND-S Algorithm

1. What is the FIND-S algorithm?

A concept learning algorithm that finds the most specific hypothesis.

2. What is meant by the most specific hypothesis?

Hypothesis covering positive examples only.

3. Is FIND-S supervised or unsupervised?

Supervised learning.

4. What are positive and negative training examples?

Positive satisfy target concept; negative do not.

5. Does FIND-S consider negative examples? Why?

No, it focuses only on positive examples.

6. What is hypothesis in machine learning?

A proposed explanation or model.

7. What symbol is used for “any value” in FIND-S?

Question mark (?) symbol.

8. What is concept learning?

Learning a general concept from examples.

9. What are the limitations of FIND-S algorithm?

Ignores negative examples and sensitive to noise.

10. Why is CSV file used in this experiment?

To store tabular data.

11. Which Python library is used to read CSV files?

Pandas.

12. Explain the working of FIND-S algorithm step by step.

Initialize specific hypothesis, generalize using positive examples, ignore negatives, and obtain final hypothesis.

Experiment 6: Locally Weighted Regression

1. What is Locally Weighted Regression?

Regression technique fitting data locally using weights.

2. Why is it called non-parametric?

Because it does not assume fixed model structure.

3. What is the role of weight in this algorithm?

Nearby points get higher importance.

4. How does bandwidth parameter affect prediction?

Small bandwidth gives flexible curve; large gives smoother curve.

5. Difference between linear regression and locally weighted regression?

Linear regression fits one line globally; LWR fits local curves.

6. What is regression analysis?

Technique used to predict continuous values.

7. What are the applications of regression?

Sales prediction, forecasting, trend analysis.

8. Why is visualization important in regression?

Helps understand trends and fitting.

9. What is curve fitting?

Finding a curve that best fits data.

10. Is locally weighted regression suitable for large datasets? Why?

No, because it is computationally expensive.

11. What is prediction in machine learning?

Estimating outputs for unseen data.

12. Name libraries used for regression plotting.

Matplotlib and Seaborn.

Experiment 7: Linear and Polynomial Regression

1. What is linear regression?

Regression method using straight-line relationship.

2. What is polynomial regression?

Regression using polynomial equations.

3. Difference between linear and polynomial regression?

Linear fits straight line; polynomial fits curves.

4. What is dependent variable?

Output variable being predicted.

5. What is independent variable?

Input feature used for prediction.

6. What is regression line?

Best-fit line representing relationship.

7. What is R^2 score?

Measures goodness of fit of model.

8. Why is polynomial regression used?

To model non-linear relationships.

9. What is underfitting?

Model is too simple.

10. What is overfitting?

Model learns noise and performs poorly on new data.

11. What dataset is used for linear regression in this experiment?

Usually Salary dataset.

12. Mention applications of regression.

Price prediction, forecasting, risk analysis.

Experiment 8: Decision Tree

1. What is a decision tree?

Tree-structured model for classification and regression.

2. Is decision tree supervised or unsupervised?

Supervised learning.

3. What is root node?

Top node representing first decision.

4. What is leaf node?

Final node representing output class.

5. What is entropy?

Measure of impurity or randomness.

6. What is information gain?

Reduction in entropy after split.

7. Why are decision trees easy to interpret?

Decisions are represented visually.

8. What is pruning in decision trees?

Removing unnecessary branches to avoid overfitting.

9. Difference between classification and regression tree?

Classification predicts categories; regression predicts continuous values.

10. What is precision?

Ratio of true positives to predicted positives.

11. What is recall?

Ratio of true positives to actual positives.

12. What are the advantages of decision trees?

Easy to understand and requires less preprocessing.

Experiment 9: Naive Bayes

1. What is Naive Bayes algorithm?

Probabilistic classification algorithm based on Bayes theorem.

2. Why is it called “naive”?

Because it assumes features are independent.

3. What is Bayes theorem?

Formula used to calculate conditional probability.

4. What is prior probability?

Probability before observing evidence.

5. What is posterior probability?

Updated probability after evidence.

6. What assumption does Naive Bayes make?

Features are conditionally independent.

7. Is Naive Bayes supervised or unsupervised?

Supervised learning.

8. What are the advantages of Naive Bayes?

Fast, simple, and efficient for text data.

9. What are the limitations of Naive Bayes?

Independence assumption may not hold.

10. Why is Iris dataset used in classification?

Simple and well-structured dataset.

11. What is classification accuracy?

Percentage of correctly classified instances.

12. Mention applications of Naive Bayes.

Spam filtering, sentiment analysis, document classification.

Experiment 10: K-Means Clustering

1. What is clustering?

Grouping similar data points together.

2. What is K-Means algorithm?

An unsupervised clustering algorithm.

3. Is K-Means supervised or unsupervised?

Unsupervised learning.

4. What is centroid in K-Means?

Center point of cluster.

5. What is the role of K value?

Specifies number of clusters.

6. How are clusters formed in K-Means?

By assigning points to nearest centroid iteratively.

7. What is Euclidean distance?

Shortest distance between two points.

8. What happens if K value is too large?

Too many small meaningless clusters form.

9. What is inertia in clustering?

Sum of squared distances between points and centroids.

10. Difference between classification and clustering?

Classification uses labeled data; clustering uses unlabeled data.

11. Why is visualization important in clustering?

Helps understand cluster separation.

12. Mention applications of K-Means clustering.

Customer segmentation, image compression, market analysis.